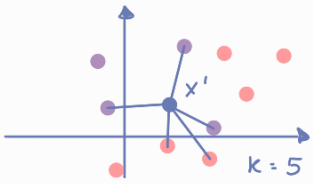
K-Nearest Neighbors (KNN)

Consider the usual classification problem $f: X \longrightarrow C$
with a dataset $D = \{(x_n, t_n)_{n=1}^{N}\}$

Classification with K-NN:

1. find the K nearest neighbors of the new instance x'
2. Assign to x the most common label among the majority of neighbors



$x' \notin D$

Likelihood of class c for new instance x':

$$P(c \mid x', D, k) = \frac{1}{k} \sum_{x_n \in N_k(x_n, D)} I(t_n = c)$$

input
dataset
number of neighbors
neighbors

in order to predict the class we will use the majority voting
● will be classified as ●

$P(● \mid ●, D, k=5) = 3/5$
$P(● \mid ●, D, k=5) = 1 - 3/5 = 2/5$

with $N_k(x_n, D)$ the k nearest points to x_n and $I(\ell) = \begin{cases} 1 & \text{iff } \ell \text{ is true} \\ 0 & \text{iff } \ell \text{ is false} \end{cases}$

When $k = 1$ this method corresponds to a Voronoi tesselation.

Increasing k brings smoother regions (reducing overfitting)

Pros { 1. No training phase

Cons { 1. Requires storing all the dataset
2. Depends on the distance function (in many cases it is not easy to define a distance function)

Kernelized KNN

In KNN everything depends on the choice of the distance function.
The distance function can be written as a kernel

$$\| x - x_n \|^2 = x^T x + x_n^T x_n + 2 x^T x_n$$

can be kernelized using a kernel $k(x, x_n)$

> When experimenting with a complex method, you should compare it with KNN (baseline) because of its simplicit